5	**(a)**		$F = 50$ Take moments about A (or D) $Fx = 20\times30 + 20\times10$ OR Take moments about B (or C) $F(30-x) + 20\times10 = 30\times30$ $x = \frac{800}{50} = 16$	**B1** **M1** **A1** **[3]**	**1.1b** **1.1a** **1.1b**	cao Forming an equation for moments about any point –allow their value of F used. Allow one incorrect distance or a missing term Allow 0.16 m if unit given	Do not allow for moment = force
5	**(b)**		Forces have a resultant moment which have a clockwise turning effect on the lamina	**E1** **E1** **[2]**	**2.4** **2.4**	Allow 'resultant moment' or 'not in equilibrium' soi Clockwise must be indicated	Do not allow for 'a resultant force' on its own